ASAD ALI HASSAN

DevSecOps Apprentice | Python · AWS security · Secure CI/CD

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PROFESSIONAL PROFILE

Part-time **DevSecOps Apprentice** at Cyber Agoge (since November 2024) building **Python** security tooling, **AWS** guardrails and **secure CI/CD** pipelines in public repositories. Recent work includes a read-only **boto3** scanner that audits AWS accounts against CIS and AWS Foundational Security Best Practices controls, tested with **pytest** and **moto**, and a **GitLab CI** pipeline with SAST, secret detection, IaC scanning, a strict control gate and **OIDC** access to AWS without static keys. Also experienced with GitHub Actions (reusable workflows, OIDC), Terraform, Kubernetes hardening, SARIF-based reporting and STRIDE threat modelling. Eight certifications including **AWS Certified Cloud Practitioner**, CompTIA Security+ and HashiCorp Terraform Associate.

Over four years in full-time manufacturing quality control (Sertec, May 2022 to present) built discipline in standards, defect prevention, traceability and auditability, which maps directly to severity-based security gates and compliance-oriented workflows.

CORE SKILLS

Python & testing	Python 3.12, boto3, pytest, pytest-cov, moto (mocked AWS), Ruff, mypy, Bandit, pip-audit; CLI tooling with table, JSON and SARIF output
AWS & cloud security	IAM least privilege, S3 Block Public Access, CloudTrail, EBS encryption, security groups, AWS Config, SCPs, KMS, STS web identity, CloudFront, WAFv2, VPC/EKS; CIS AWS Foundations Benchmark and AWS FSBP control mapping; Cloudflare Pages, Wrangler
CI/CD & supply chain	GitLab CI/CD (staged pipelines, OIDC id_tokens, SAST, secret detection, IaC SAST, JUnit/coverage/Code Quality reports), GitHub Actions (reusable workflows, OIDC, branch protection), SARIF / code scanning, Gitleaks, TruffleHog, Semgrep, CodeQL, Trivy, Checkov, KICS, ZAP, Dependabot, Dependency Review, CycloneDX SBOM
Infrastructure as code	Terraform (reusable modules, terraform test with mocked providers, Cloudflare and AWS providers), Checkov and Trivy IaC scanning, documented policy exceptions
Containers & runtime	Docker (non-root, digest-pinned images), Kubernetes (ingress/TLS, NetworkPolicy, pod hardening, probes, limits), Falco, Prometheus, Grafana dashboards as code
Application stack	JavaScript, TypeScript, Node.js, Express, React, Vite, REST APIs, Supabase (RLS/JWT awareness); Linux, Git
Security practice	STRIDE threat modelling, vulnerability triage, severity-based pipeline gates, secure SDLC documentation, risk-based controls

PROFESSIONAL EXPERIENCE

DevSecOps Apprentice — Cyber Agoge

Nov 2024 – present

Part-time apprenticeship, United Kingdom; undertaken alongside full-time Quality Checker role at Sertec.

- Implement security controls across the software development lifecycle, emphasising early detection and prevention of vulnerabilities.
- Support secure CI/CD pipelines with integrated **SAST**, **secrets scanning** and **container/image scanning**; align local hooks with CI where applicable (Husky / Gitleaks patterns on public demos).
- Contribute to hardening of production-style infrastructure, including containers and Kubernetes configurations.
- Apply **STRIDE** threat modelling to identify risks and map mitigations to technical controls documented in-repo.
- Collaborate with engineering stakeholders to improve secure coding practices and pipeline reliability; treat documentation and case studies as first-class deliverables for auditability.

Quality Checker — Sertec Group Ltd (Sertec Auto Structures)

May 2022 – present

Full-time, United Kingdom.

- Inspected automotive components against strict quality standards, identifying defects (e.g. splits, distortion, surface issues).
- Reduced risk of defective output through consistent inspection and escalation of recurring quality issues.
- Maintained high accuracy under time-constrained production conditions.
- Built attention to detail, traceability and compliance discipline, directly applicable to security validation, evidence retention and gate-based workflows.

Warehouse Operative — GXO, Tamworth, UK

Mar 2021 – Apr 2022

- Picked, packed and processed stock using handheld scanning systems, maintaining stock accuracy and following safety procedures; reported discrepancies to supervisors.

Machine Operator — Vesteraalens AS (fish factory), Sortland, Norway — Sep 2018 – Dec 2020

- Operated and monitored seafood processing machinery on the production line, following hygiene and safety standards and reporting machine faults.

Teaching Assistant — Primary school, Mo i Rana, Norway — 2016 – 2018

- Supported teachers in classes 1 and 2 (Year 1–2), helping children with reading, writing and classroom activities with patience and clear communication.

SELECTED PROJECTS

aws-guardrail-scanner | github.com/asadyare/aws-guardrail-scanner

Python 3.12 · boto3 · pytest · moto · SARIF · Docker · GitHub Actions

- Built a **read-only Python CLI** that audits an AWS account with **boto3** for S3, IAM, security group, CloudTrail and EBS misconfigurations, each check mapped to **CIS AWS Foundations Benchmark v3.0.0** and **AWS FSBP** controls with remediation guidance.
- Outputs **table, JSON and SARIF 2.1.0** (for GitHub code scanning); a **--fail-on** severity threshold lets CI block a release on high or critical findings.
- **39 pytest tests** using **moto** (no live AWS calls) at **~88% coverage**, with an 80% minimum enforced; CI runs Ruff, mypy, Bandit and pip-audit.
- Ships a **least-privilege IAM policy** (iam/scanner-policy.json) limited to the read calls the checks make, and a non-root Docker image pinned by digest.

aws-guardrail-pipeline | gitlab.com/asadyare/aws-guardrail-pipeline

GitLab CI/CD · Terraform · Python · Checkov · Trivy · Semgrep · OIDC

- Wrote **Terraform guardrail modules** for an AWS account baseline (S3 Block Public Access, EBS encryption by default, multi-Region CloudTrail with KMS, IAM password policy, AWS Config rules, tamper-deny SCP), tested with terraform test.
- Built a **Python guardrail-compliance checker** that evaluates Terraform plan JSON against the baseline and emits JUnit and GitLab Code Quality reports for merge requests.
- Designed a **staged GitLab CI pipeline** with security gates: SAST (Semgrep), secret detection, IaC scanning (Checkov, Trivy, KICS), Bandit, pip-audit and a strict **control-gate** job; images pinned by digest.
- Added a manual **scan-aws** job that uses **GitLab OIDC id_tokens** and STS web identity (no static AWS keys) to run the scanner pinned to a commit SHA.
- Debugged a control-gate failure: files declared only under artifacts:reports were not passed to later jobs; fixed by also listing them under artifacts:paths, keeping the gate strict.

Wider DevSecOps portfolio

- **Secure banking app demo** – React/TypeScript/Supabase SPA; Terraform on AWS (S3, CloudFront, WAFv2); OIDC deploy; CodeQL, Checkov, ZAP; incident-style case study.
- **Healthcare apps monorepo** – Express + React services; Terraform (VPC, private EKS API, KMS, flow logs); matrix image builds with Checkov, Trivy and kubectl dry-run gates; STRIDE and HIPAA-oriented design notes (not a compliance attestation).
- **Shared security CI/CD** – Reusable GitHub Actions: Gitleaks, Semgrep, npm audit, Docker build, Trivy image scan, SARIF integration.
- **Kubernetes security** – Hardened manifests, ingress/TLS, NetworkPolicy, Falco runtime detection, Grafana as code.
- **Daily security automation** – Scheduled scans, SBOM, SARIF, GitHub Issues as an audit trail for critical findings.
- **Threat modelling** – STRIDE, trust boundaries, mitigations mapped to CI, Kubernetes and cloud controls.
- **Portfolio frontend** – React (Vite) on Cloudflare Pages; Terraform for DNS/zone; Pages Functions for metrics.

Architecture pack (PDF): DevSecOps Architecture Pack

CERTIFICATIONS

AWS Certified Cloud Practitioner	Apr 2025 – Apr 2028	**CompTIA Security+**	Mar 2025 – Mar 2028
HashiCorp Terraform Associate (003)	Apr 2025 – Apr 2027	**GitHub Actions Certification (Microsoft Learn)**	Oct 2025 – Oct 2027
ISC2 Certified in Cybersecurity (CC)	Jun 2024 – Jun 2027	**DevSecOps Certified (Cyber Agoge)**	Dec 2024
Google Cybersecurity Specialization (Coursera)	Jun 2024	**Mastercard Cybersecurity Job Simulation (Forage)**	Jun 2024

EDUCATION

Secondary education – Narvik videregående skole, avdeling Oscarsborg, 2014–2016

Self-directed learning – cloud security and DevSecOps through hands-on labs and project-based work on GitHub and GitLab (projects above).

ADDITIONAL INFORMATION

Languages: English – professional working proficiency

Work authorisation: Eligible to work in the United Kingdom